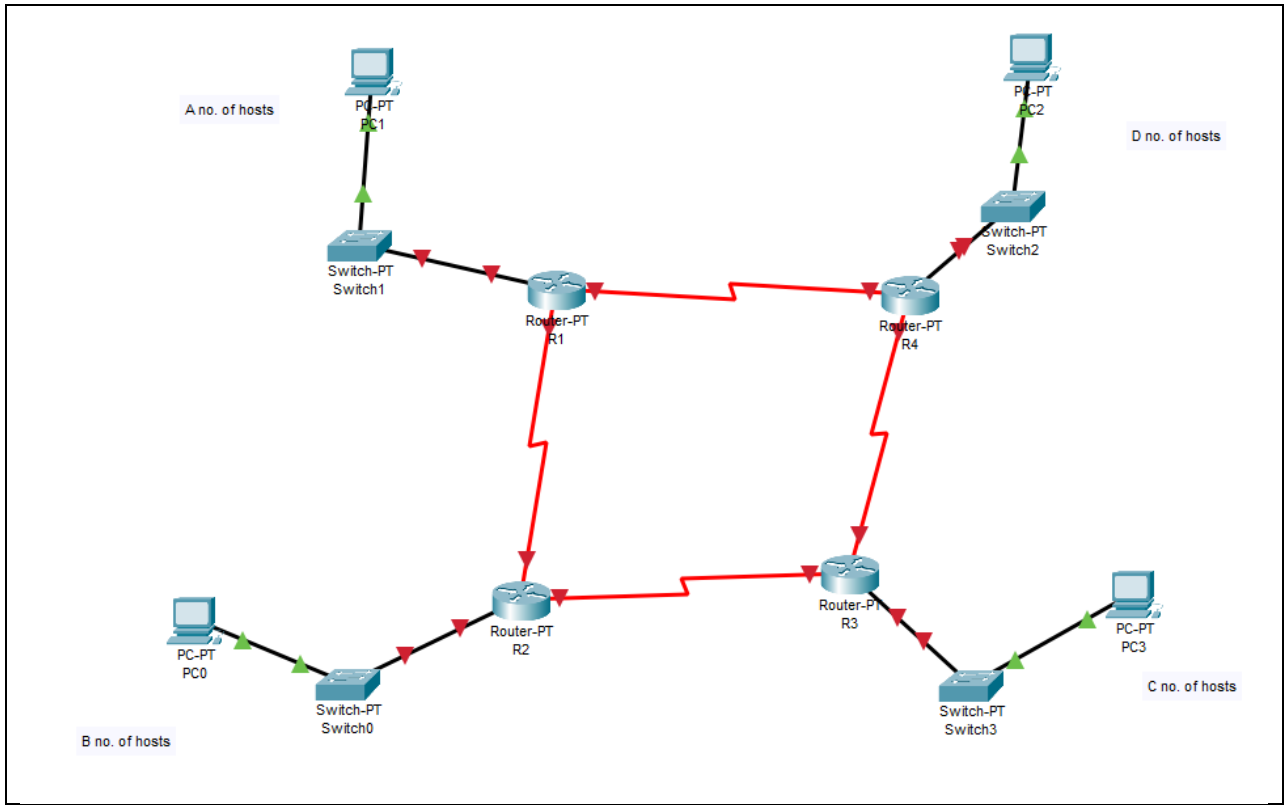
CN LAB Assignment
Batch -79AB

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1. Design the following topology in packet tracer using Static, RIP, OSPF routing (separately).



2. Fill up the following table 1, 2, 3.

Table 1: Subnet Information

Subnet Number	Subnet Address and Subnet Mask	First Host Address	Last Host Address	Gateway Address
1 (3000 Hosts)	10.128.0.0/20 255.255.240.0	10.128.0.1/20	10.128.15.254/20	10.128.0.1/20
2 (300 Hosts)	10.128.16.0/23 255.255.254.0	10.128.16.1/23	10.128.17.254/23	10.128.16.1/23
3 (30 Hosts)	10.128.18.0/27 255.255.255.224	10.128.18.1/27	10.128.18.30/27	10.128.18.1/27
4 (30 Hosts)	10.128.18.32/27 255.255.255.224	10.128.18.33/27	10.128.18.62/27	10.128.18.33/27
5 (2 Hosts)	10.128.18.64/30 255.255.255.252	10.128.18.65/30	10.128.18.66/30	N/A
6 (2 Hosts)	10.128.18.68/30 255.255.255.252	10.128.18.69/30	10.128.18.70/30	N/A
7 (2 Hosts)	10.128.18.72/30 255.255.255.252	10.128.18.73/30	10.128.18.74/30	N/A
8 (2 Hosts)	10.128.18.76/30 255.255.255.252	10.128.18.77/30	10.128.18.78/30	N/A

Table 2: Addressing Table

Device	Interface	IP address	Subnet Mask
R1	Fa0/0	10.128.0.1/20	255.255.240.0
	Se2/0	10.128.18.65/30	255.255.255.252
	Se3/0	10.128.18.77/30	255.255.255.252
R2	Fa0/0	10.128.16.1/23	255.255.254.0
	Se2/0	10.128.18.66/30	255.255.255.252
	Se3/0	10.128.18.69/30	255.255.255.252
R3	Fa0/0	10.128.18.1/27	255.255.255.224
	Se2/0	10.128.18.73/30	255.255.255.252
	Se3/0	10.128.18.70/30	255.255.255.252
R4	Fa0/0	10.128.18.33/27	255.255.255.224
	Se2/0	10.128.18.74/30	255.255.255.252
	Se3/0	10.128.18.78/30	255.255.255.252

Table 3: Static Routing Table

Router	Unknown Network Address	Mask Address	Next Hop
R1	**10.128.16.0/23**	**255.255.254.0**	**10.128.18.78/30**
	10.128.18.32/27	**255.255.255.224**	**10.128.18.78/30**
	10.128.18.0/27	**255.255.255.224**	**10.128.18.78/30**
	10.128.18.72/30	**255.255.255.252**	**10.128.18.78/30**
	10.128.18.68/30	**255.255.255.252**	**10.128.18.78/30**
R2	**10.128.0.0/20**	**255.255.240.0**	**10.128.18.65/30**
	10.128.18.0/27	**255.255.255.224**	**10.128.18.65/30**
	10.128.18.32/27	**255.255.255.224**	**10.128.18.65/30**
	10.128.18.72/30	**255.255.255.252**	**10.128.18.65/30**
	10.128.18.76/30	**255.255.255.252**	**10.128.18.65/30**
R3	**10.128.18.32/27**	**255.255.255.224**	**10.128.18.69/30**
	10.128.16.0/23	**255.255.254.0**	**10.128.18.69/30**
	10.128.0.0/20	**255.255.240.0**	**10.128.18.69/30**
	10.128.18.76/30	**255.255.255.252**	**10.128.18.69/30**
	10.128.18.64/30	**255.255.255.252**	**10.128.18.69/30**
R4	**10.128.0.0/20**	**255.255.240.0**	**10.128.18.73/30**
	10.128.18.0/27	**255.255.255.224**	**10.128.18.73/30**
	10.128.16.0/23	**255.255.254.0**	**10.128.18.73/30**
	10.128.18.64/30	**255.255.255.252**	**10.128.18.73/30**
	10.128.18.68/30	**255.255.255.252**	**10.128.18.73/30**

Table 4: RIP Routing Table

Router	Neighbor Network Address
R1	**10.128.0.0/20**
	10.128.18.64/30
	10.128.18.76/30
R2	**10.128.16.0/23**
	10.128.18.64/30
	10.128.18.68/30
R3	**10.128.18.0/27**
	10.128.18.68/30
	10.128.18.72/30
R4	**10.128.18.32/27**
	10.128.18.72/30
	10.128.18.76/30

Table 5: OSPF Routing Table

Router	Neighbor Network Address	Wildcard Mask
R1	**10.128.0.0/20**	**0.0.15.255**
	10.128.18.64/30	**0.0.0.3**
	10.128.18.76/30	**0.0.0.3**
R2	**10.128.18.64/30**	**0.0.0.3**
	10.128.18.68/30	**0.0.0.3**
	10.128.16.0/23	**0.0.1.255**
R3	**10.128.18.68/30**	**0.0.0.3**
	10.128.18.0/27	**0.0.0.31**
	10.128.18.72/30	**0.0.0.3**
R4	**10.128.18.72/30**	**0.0.0.3**
	10.128.18.76/30	**0.0.0.3**
	10.128.18.32/27	**0.0.0.31**

4. Address space according to your Student ID.

Your ID	A No. of Hosts	B No. of Hosts	C No. of Hosts	D No. of Hosts	Address Space
CSE 078 08384	1000	2000	3000	1000	172.0.0.0/16
CSE 079 08430	100	200	300	200	172.0.128.0/17
CSE 079 08431	500	1000	600	500	172.0.192.0/18
CSE 079 08432	10	20	200	100	170.0.0.0/8
CSE 079 08434	500	1000	600	50	10.224.0.0/11
CSE 079 08435	50	100	700	50	10.240.0.0/12
CSE 079 08437	30	500	4000	30	120.0.0.0/8
CSE 079 08438	2000	200	20	20	5.0.0.0/8
CSE 079 08439	3000	300	30	30	10.128.0.0/9
CSE 079 08441	5	10	150	150	192.0.0.0/10
CSE 079 08445	35	150	250	35	192.192.0.0/10
CSE 079 08446	2000	3000	4000	2000	10.0.0.0/8

CSE 079 08451	1000	100	10	10	10.0.0.0/8
CSE 079 08454	1000	2000	5	5	165.16.0.0/16
CSE 079 08455	5	16	8	16	15.0.0.0/8
CSE 079 08458	150	25	756	15	16.128.0.0/9
CSE 079 08460	700	800	900	800	16.192.0.0/10
CSE 079 08480	2000	2000	4000	1000	10.128.0.0/9
CSE 079 08452	512	10	1000	500	10.224.0.0/11
CSE 079 08461	323	123	423	300	170.0.0.0/8
CSE 079 08470	15	515	5000	15	3.0.0.0/8
CSE 079 08472	5	10	30	10	192.160.15.0/24
CSE 079 08476	100	10	25	10	192.160.254.0/23
CSE 079 08478	1500	1800	800	800	4.0.0.0/9
CSE 079 08479	1800	2000	10	800	5.0.0.0/8
CSE 079 08483	3000	4000	100	110	18.0.0.0/8
CSE 079 08484	3000	2000	5000	140	175.16.0.0/16
CSE 079 08489	700	800	80	16	15.0.0.0/8
CSE 079 08496	150	25	756	150	16.128.0.0/9
CSE 079 08500	700	800	900	800	15.192.0.0/10
CSE 079 08501	1800	18	3400	3000	18.0.0.0/8
CSE 079 08504	8522	58	5	2000	10.128.0.0/9
CSE 079 08507	54	54	4522	1000	165.16.0.0/16
CSE 079 08508	12	8999	8999	500	5.0.0.0/8
If ID not in the list	2000	3000	60	2000	10.0.0.0/8